



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 3: Consumer Theory

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ICS2523 Microeconomics

Lecture 3

Last lecture we

- Gave the **First Assignment**
- Discussed the difference between Microeconomics and Macroeconomics
- Defined Economic Models
- Compared Partial Equilibrium Models vs. General Equilibrium Models

In today's lecture, we will

- Start with **Consumer Theory**
 - Preferences and Utility
 - Indifference Curves

Consumer Theory

Chapter 2

Consumer Theory

We will study how consumers behave in order to explain how **individuals make choices**:

- **Conceptualize utility** in a mathematical way → “indifference curves”
- **Understand utility maximization** → theory of choice
- Investigate **how individuals may respond** to changes in their circumstances (changes in income, prices)

Steps:

1. **Understand individual's preferences**
2. Know individual's budget constraint
3. Final choice

Preferences and Utility

Consumer Theory

1. Preferences and Utility

Consumption Bundle:

List of quantities (x_i) of one or more goods $x = \begin{pmatrix} x_1 \\ x_2 \\ \nwarrow \\ x_n \end{pmatrix}$

Where x_i is the number of units of good i in the bundle.

(Bundle: a number of things tied or wrapped together)

Set $\mathbf{X}$ its individual's consumption set and we will assume that it is a closed and convex set.

1. Preferences and Utility

Bundle:



<https://forum.wordreference.com/threads/bundle.3044148/>

Basket (Canasta):



https://es.123rf.com/photo_81520886_imagen-de-la-historieta-del-icno-de-la-cesta-s%C3%ADmbolo-de-canasta.html

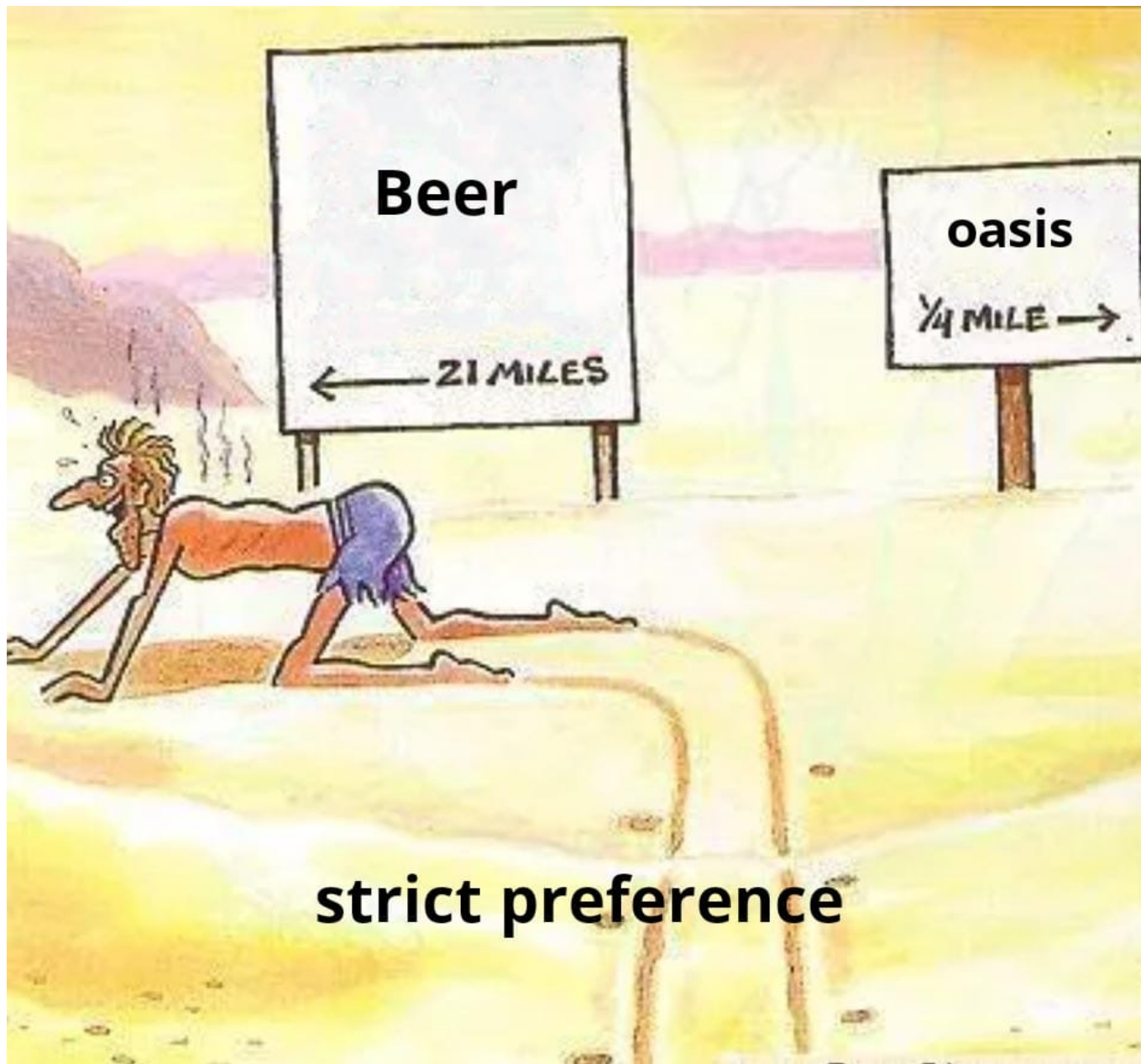
1. Preferences and Utility

Consumer is assumed to have preferences on the consumption bundles in $\mathbf{X}$:

$x \geq y \rightarrow$ consumer thinks that bundle x is at least as good as bundle y (**weak preference**)

$x > y \rightarrow$ consumer prefers bundle x from bundle y (**strict preference**)

$x \sim y \rightarrow$ consumer is **indifferent**



strict preference

1. Preferences and Utility

Axioms of rational choice

1. Completeness:

$$\forall x, y \in X: x \geq y \vee y \geq x$$

2. Transitivity:

$$\forall x, y, z \in X: x \geq y \wedge y \geq z \Rightarrow x \geq z$$

(More is better than less)

3. Continuity:

For all y in X , the sets $\{x: x \geq y\}$ and $\{x: x \leq y\}$ are closed sets. It follows that $\{x: x > y\}$ and $\{x: x < y\}$ are open sets

2. Utility Function

Given the three axioms of rational choice (Completeness, transitivity, and continuity) **people are able to rank** all possible situations from the least desirable to the most.

If an individual has to choose among n consumption goods $(x_1, x_2, \dots, x_n)$, we will assume that **her ranking of these goods can be represented by a utility function** of the form:

$$U = U(x_1, x_2, \dots, x_n)$$

2. Utility Function

$$U = U(x_1, x_2, \dots, x_n)$$

$x_1, x_2, \dots, x_n$ are the quantities of each of n goods that might be consumed in a period.

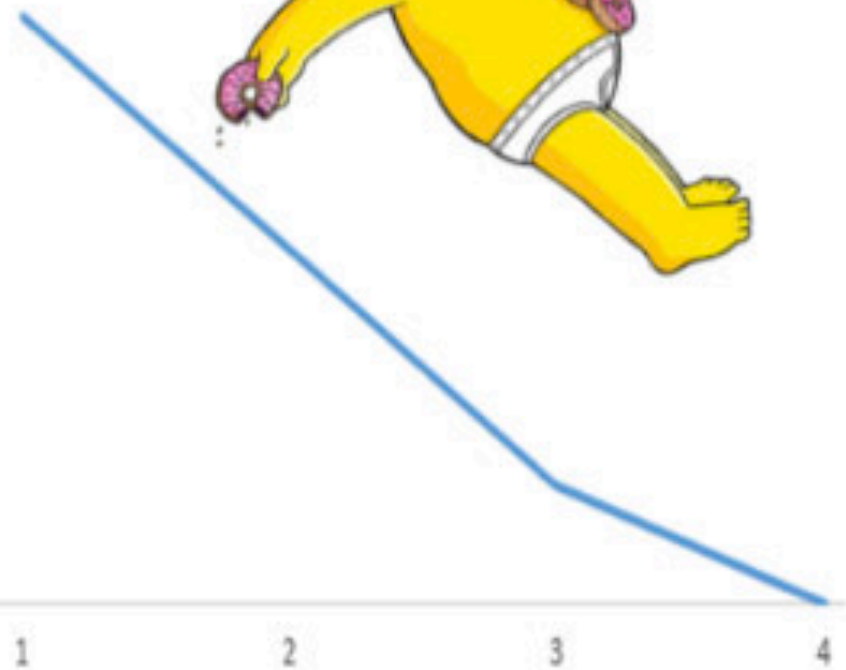
More is better than less $\rightarrow \frac{\partial U}{\partial x_i} > 0$ or $U'(x_i) > 0$

Marginal Utility (MU): is the first derivative, how utility changes with an extra unit of x_i .

**Homero thinks that
the more donuts
you've eaten, the more
utility it generates you
yo eat one more**



**Marge thinks he is
wrong and challenges
him to test**

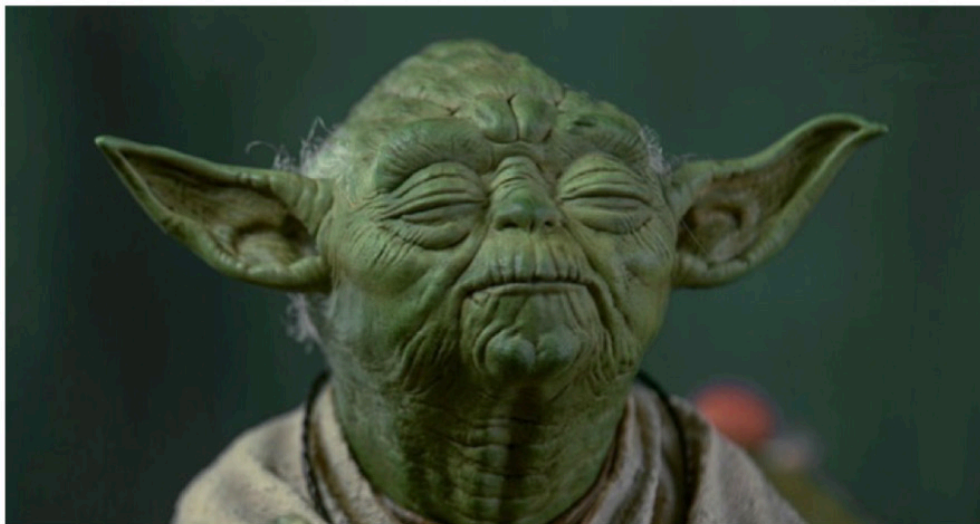


Credits: Luca Figari (2020)



Me buying 10 hot dogs thinking “the more the better”

Author:
Florencia
Kobus
(2021)



Me after eating the third hot dog
thinking about the decreasing of the
MRS

Activity 1

How is $U''(x_i) = ?$

- a) Positive
- b) Negative
- c) Zero



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<https://app.wooclap.com/DPEXTD?from=instruction-slide>

2. Utility Function

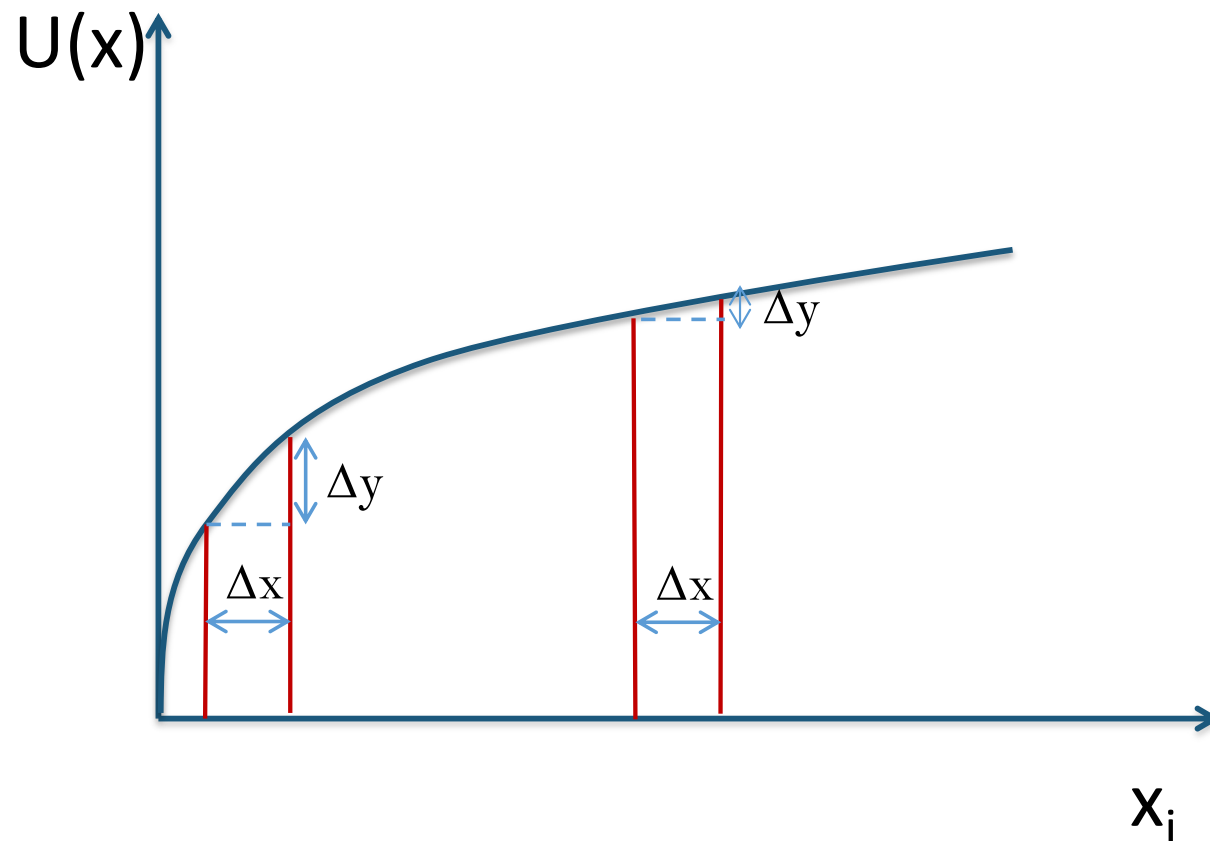
Assumption 1: $U(x)$ is doubly differentiable, increasing and strictly concave

Assumption 2: A level zero exists (x_0)

→ There is an optimum solution x^* to the economic problem of consumer utility maximization

2. Utility Function

How is $U''(x_i) = ?$ $U''(x_i) < 0$
Marginal Utility **decreases**

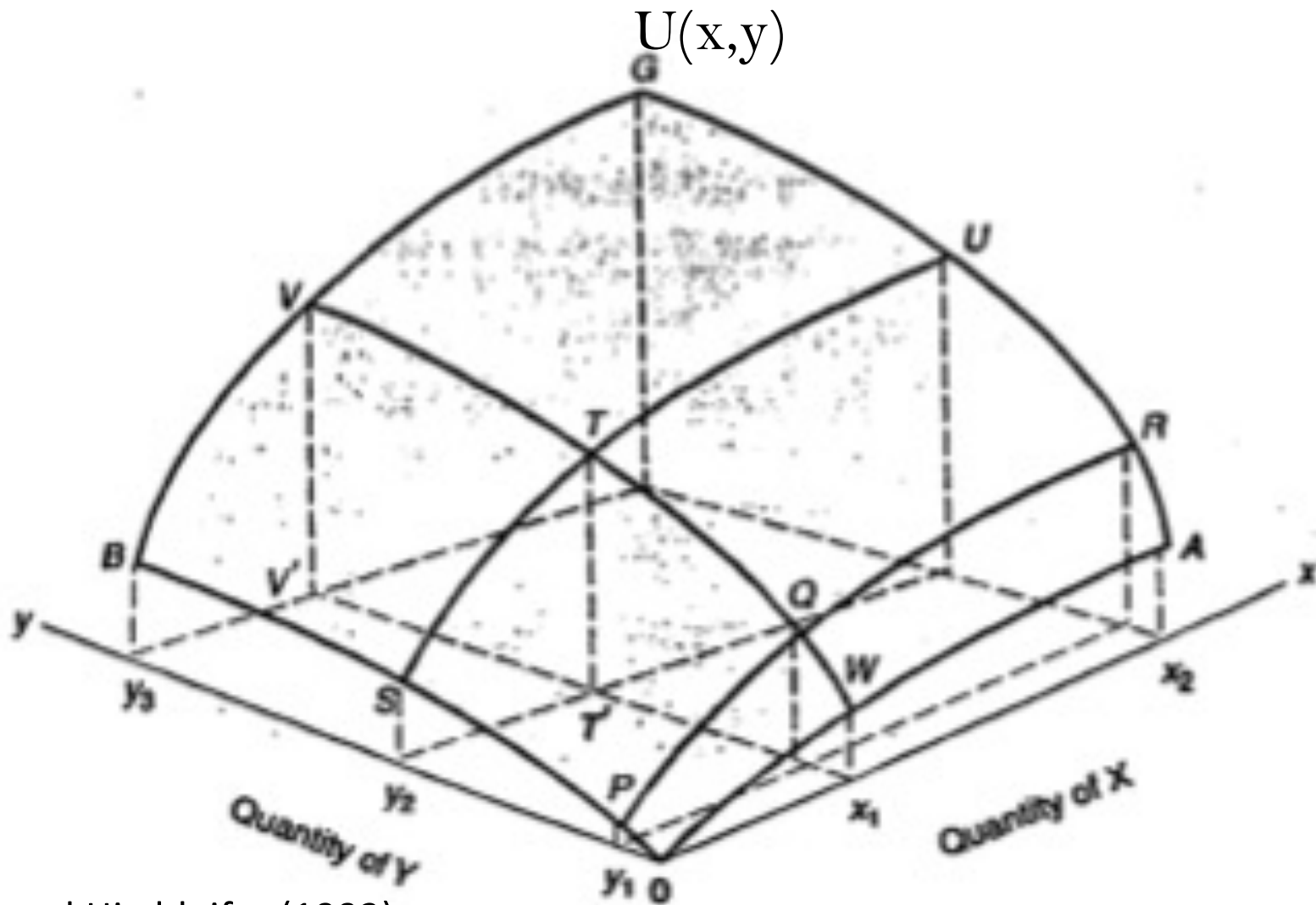


From a cardinal utility function to an ordinal utility function

We will think in **3-D** now, and our z axis will be $U(x,y)$

Imagine (ideally draw) how a cardinal total utility function of two goods should be

From a cardinal utility function to an ordinal utility function



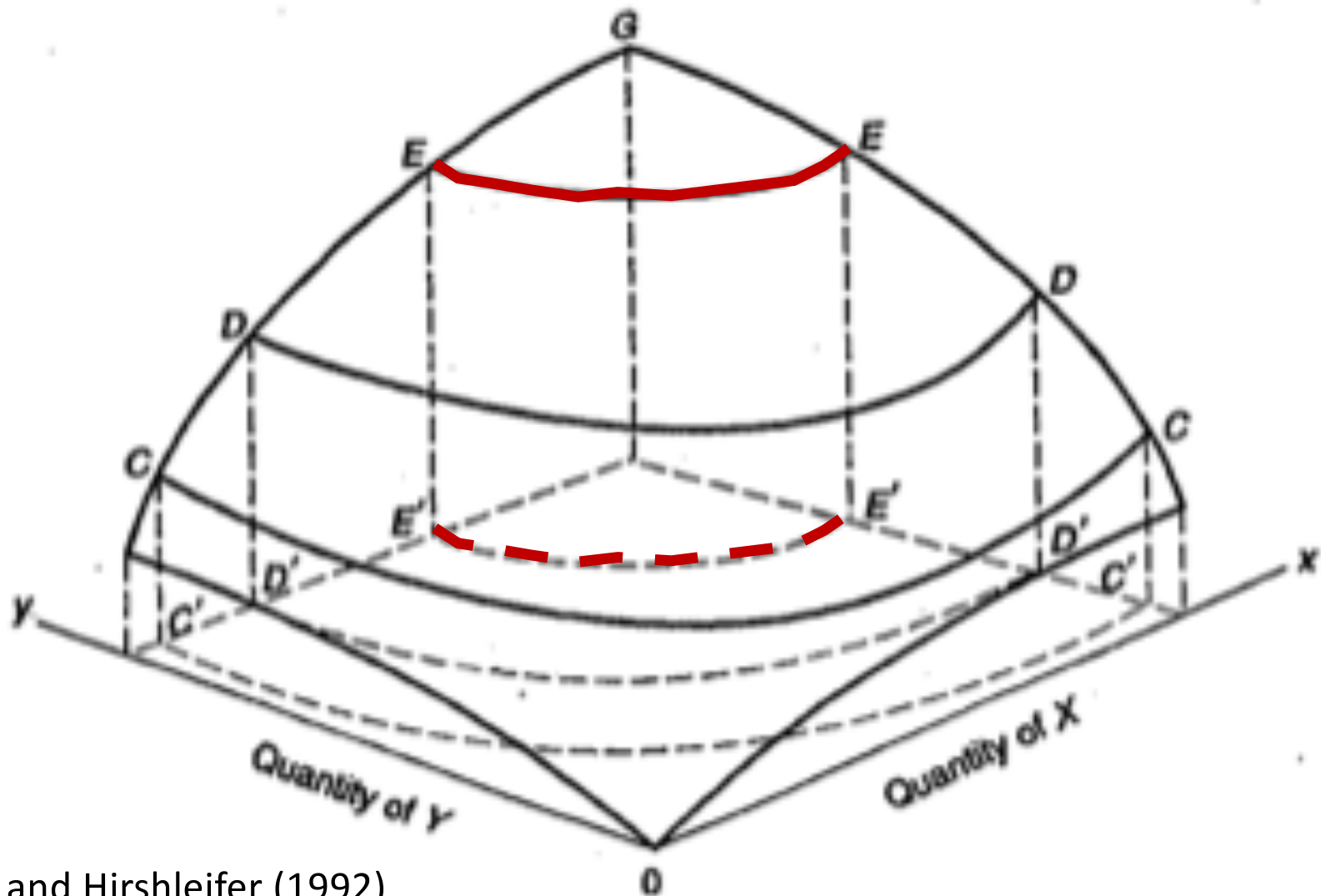
From a cardinal utility function to an ordinal utility function

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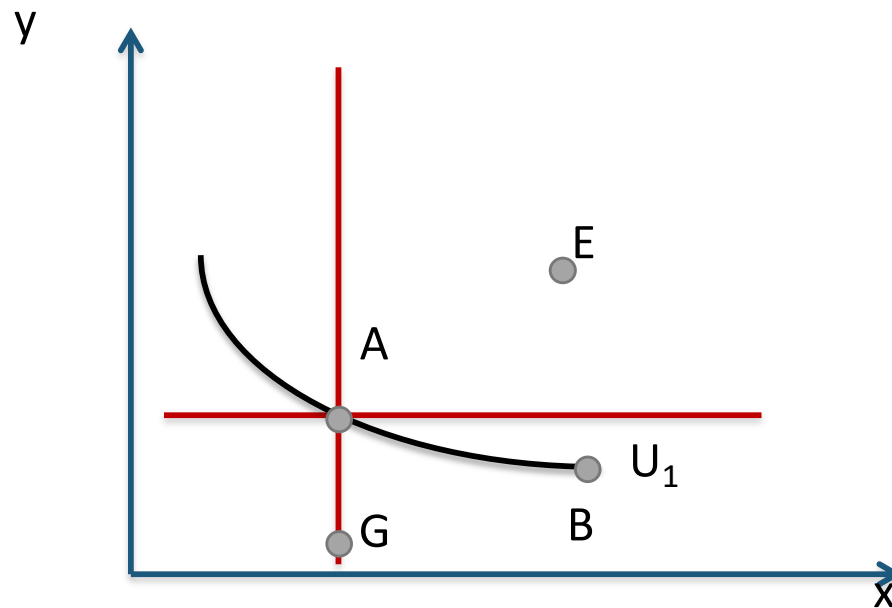
Make a vertical cut on the z-axis (parallel to XY plane-e.g. The same utility value) and think how this cuts are seen in the plane XY. What are you seeing? What does this represent?

From a cardinal utility function to an ordinal utility function



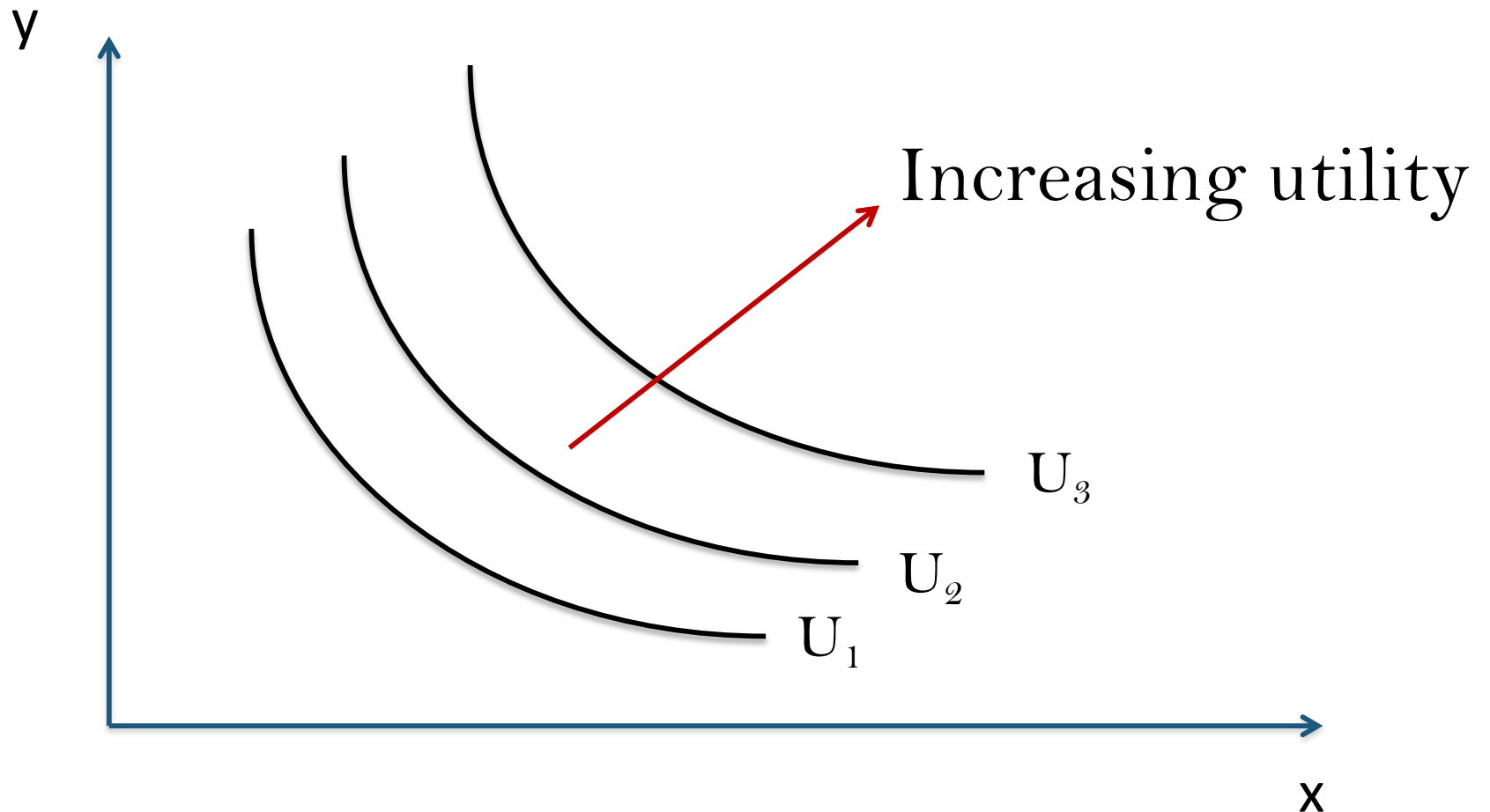
3. Indifference curves

Indifference curves: represents all those combinations of x and y from which the individual derives the same utility



3. Indifference curves

Indifference curve map: $U_1 < U_2 < U_3$



Activity 2

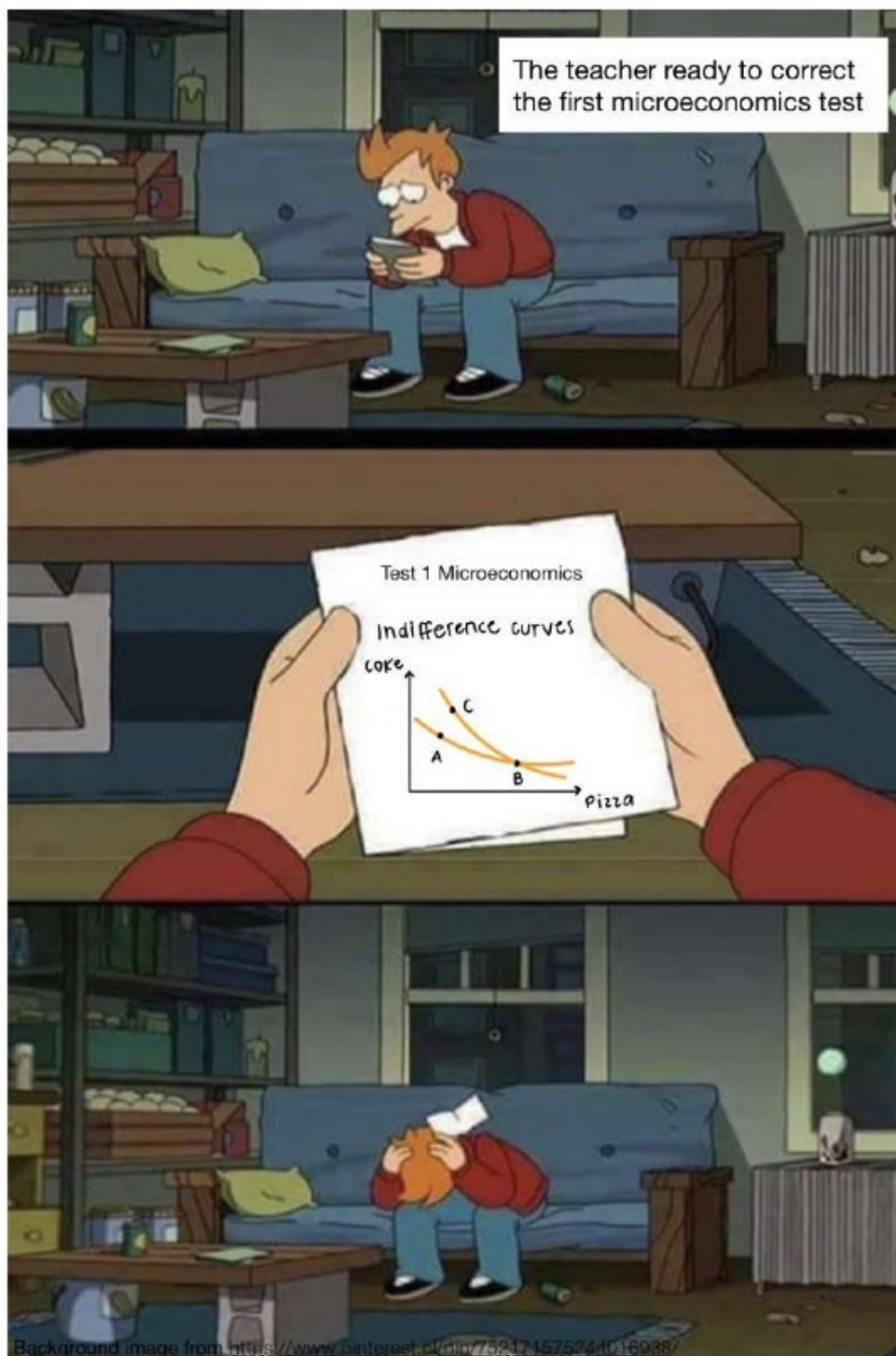
Can indifference curves intersect?

- a) Yes
- b) No



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Author: Barbara Kullmer (2023)

Marginal Rate of Substitution (MRS)

Marginal Rate of Substitution (MRS):

MRS is the quantity of certain good (y) that a consumer is **willing to give up** to obtain an extra unit of another good (x)

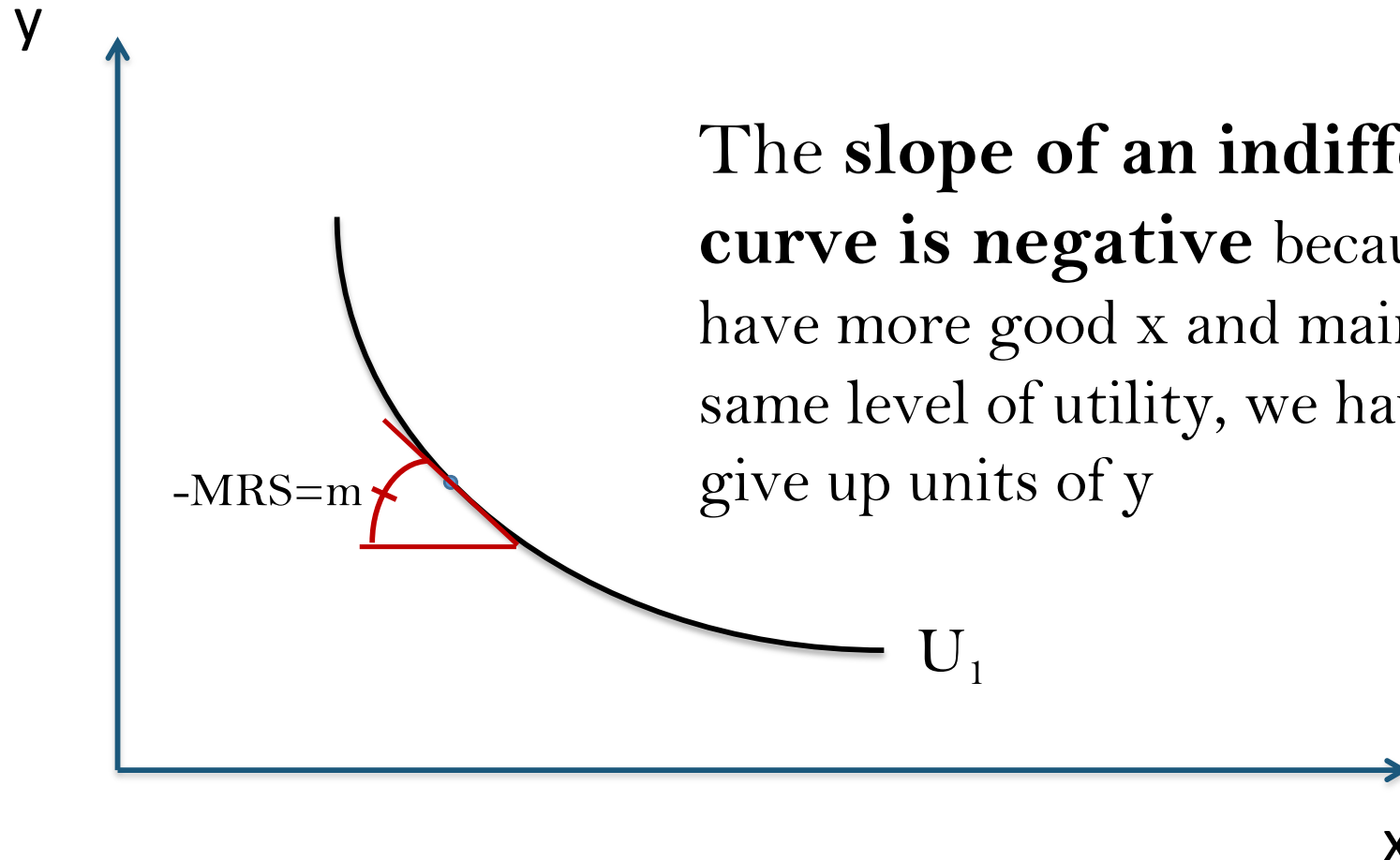
The **negative of the slope** (*pendiente*) of an **indifference curve** (U_1) at some point is the MRS at that point

$$MRS = - \left. \frac{dy}{dx} \right|_{U=U_1} = \frac{\partial U / \partial x}{\partial U / \partial y} = \frac{MU_x}{MU_y}$$

MU_x : is the marginal utility of x and the extra satisfaction obtained by consuming one more unit of x .

Marginal Rate of Substitution (MRS)

Marginal Rate of Substitution (MRS):



The **slope of an indifference curve is negative** because to have more good x and maintain the same level of utility, we have to give up units of y

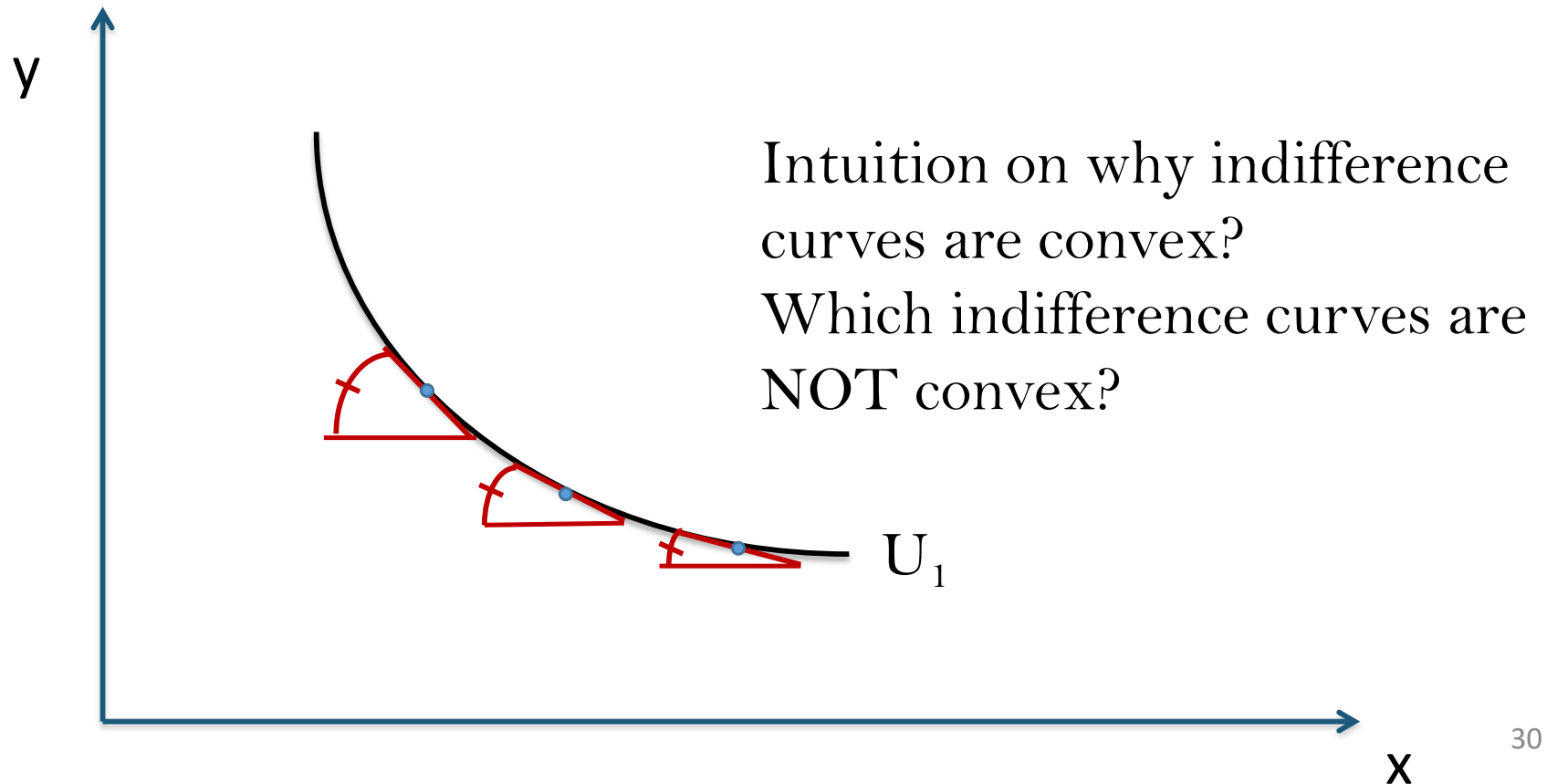
The negative of the slope of an indifference curve is defined to be the marginal rate of substitution (MRS).



Author: Benjamín Sepúlveda (2023)

Marginal Rate of Substitution (MRS)

Indifference curves are **convex** because MRS is decreasing as we move along the curve.



Marginal Rate of Substitution (MRS)

Mathematical derivation for MRS:

If the utility a person receives from two goods is represented by $U(x,y)$, we can write the total differential of this function as:

$$dU = \frac{\partial U}{\partial x} \cdot dx + \frac{\partial U}{\partial y} \cdot dy$$

This expression equals to zero along any particular indifference curve (since it is the same level of utility)

$$dU = \frac{\partial U}{\partial x} \cdot dx + \frac{\partial U}{\partial y} \cdot dy = 0$$
$$MRS = - \frac{dy}{dx} \Big|_{U=\text{constant}} = \frac{\partial U / \partial x}{\partial U / \partial y} = \frac{MU_x}{MU_y}$$

Marginal Rate of Substitution (MRS)

¿How does the MRS change in the many-good case?

Suppose Utility is a function given by $U(x_1, x_2, \dots, x_n)$ its total differential will be:

$$dU = \frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2 + \dots + \frac{\partial U}{\partial x_n} dx_n$$

Setting $dU = 0$ and holding constant quantities of all goods except for x_i and x_j (ceteris paribus):

$$dU = 0 = \frac{\partial U}{\partial x_i} dx_i + \frac{\partial U}{\partial x_j} dx_j$$

$$MRS_{x_i \text{ for } x_j} = - \frac{dx_j}{dx_i} = \frac{\partial U / \partial x_i}{\partial U / \partial x_j}$$

3. Indifference Curves

The shape of an indifference curve reflects the **desire of an individual to substitute one good for another one.**

They should be:

1. With negative slope
2. Convex to the origin
3. Indifferent curves cannot intersect
4. An indifference curve passes through each point in commodity space

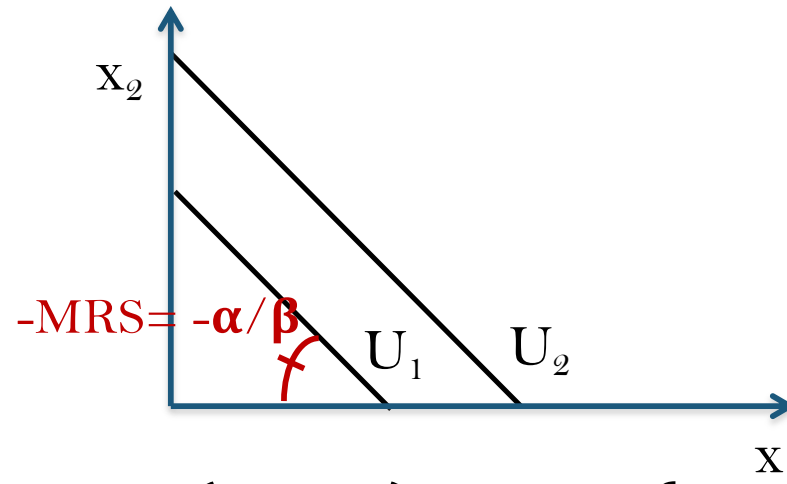
3. Indifference Curves

Examples of Utility Functions:

a) Perfect Substitutes: $U(x_1, x_2) = \alpha x_1 + \beta x_2$

Examples?

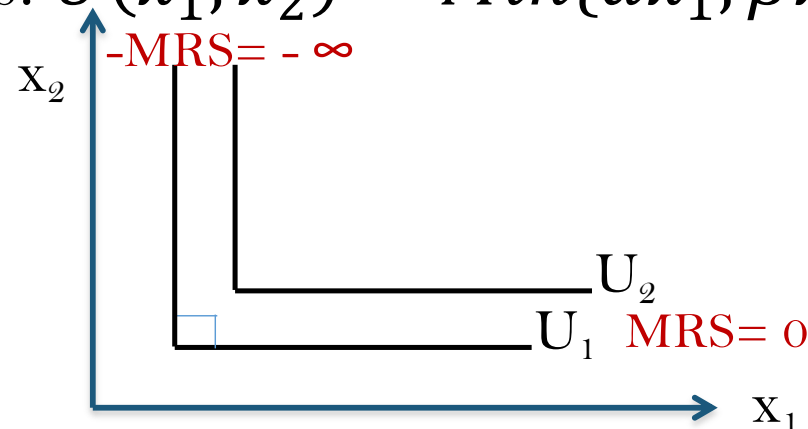
MRS is constant

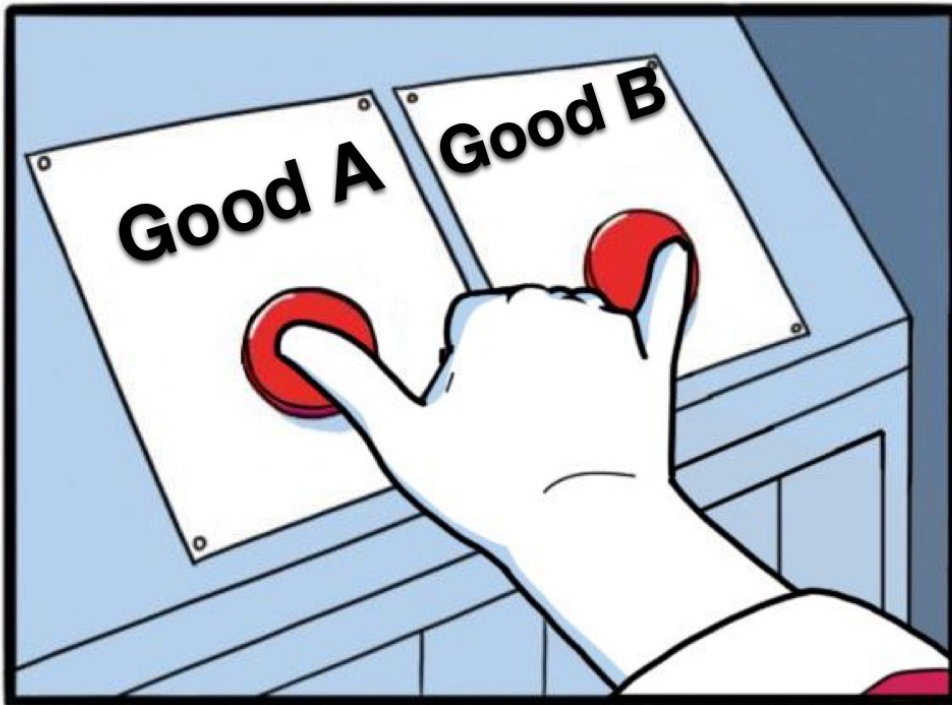


b) Perfect Complements: $U(x_1, x_2) = \text{Min}\{\alpha x_1, \beta x_2\}$

Examples?

MRS = 0 or ∞





Author: Raimundo Salas (2023)

Made with mematic



@Petirep

+ JAKE-CLARK.TUMBLR

when Pepsi and Coke are perfect substitutes and the price of Pepsi increases

Pepsi



Coke



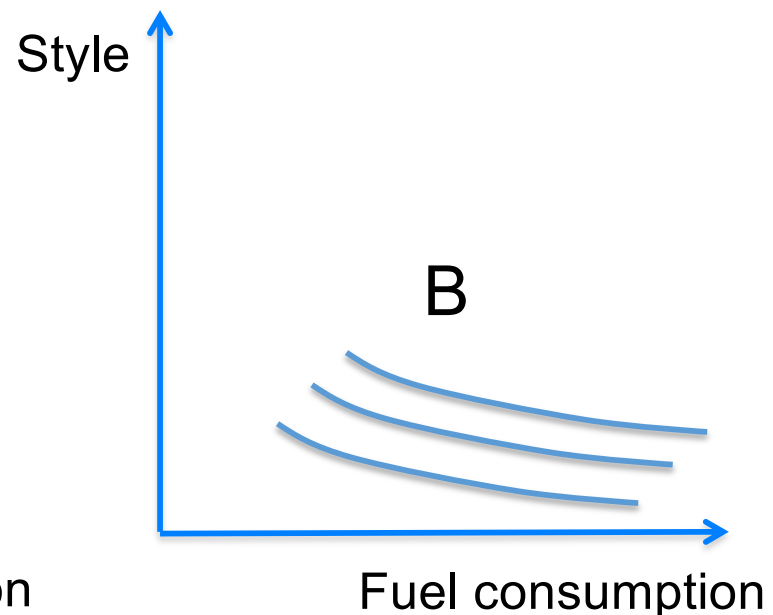
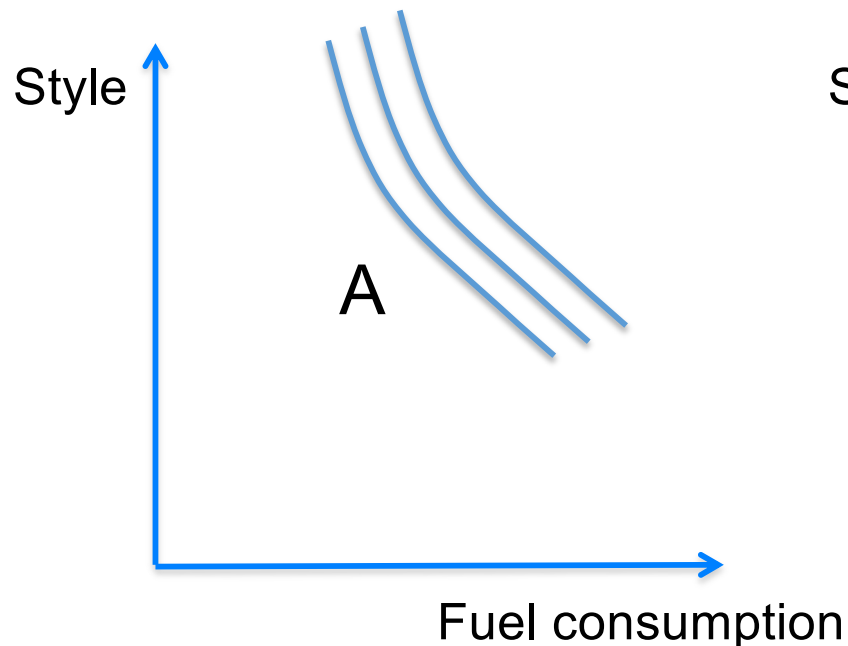
Recovered from:
<https://imgflip.com/memetemplate/444206039/Pedro-Pascal-and-Nic-Cage-car-scene>

Author: Valeria Salas (2023)

Activity 3

Imagine we have two types of people when choosing a car: one group that values fuel consumption more than **style** and another that prefers **style** over **fuel consumption**.

Look at the indifference curve maps drawn below and decide which one will be the one representing the group that values more fuel consumption (i.e. will prefer a Toyota Yaris):

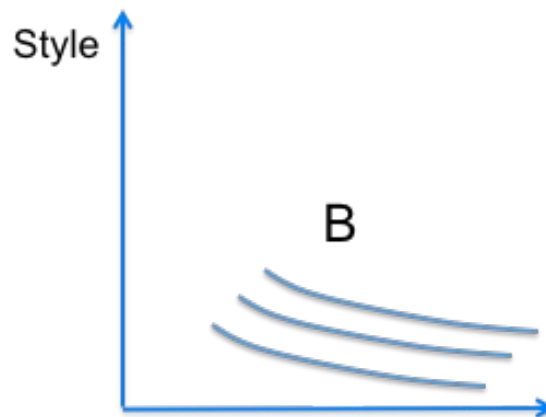
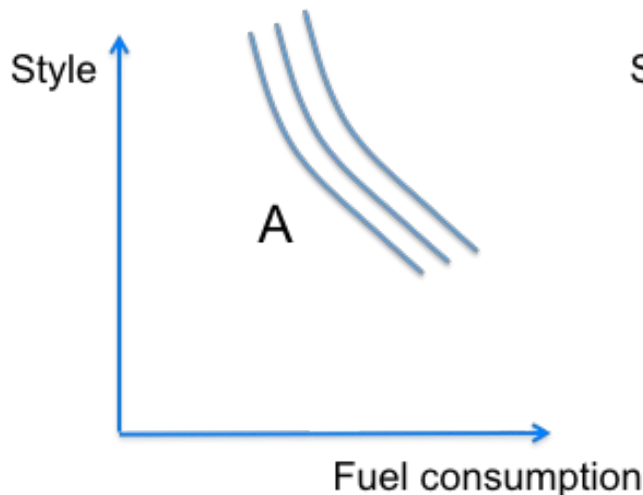


Decide which one will be the one representing the group that value more fuel consumption (i.e. Will prefer a Toyota Yaris):?

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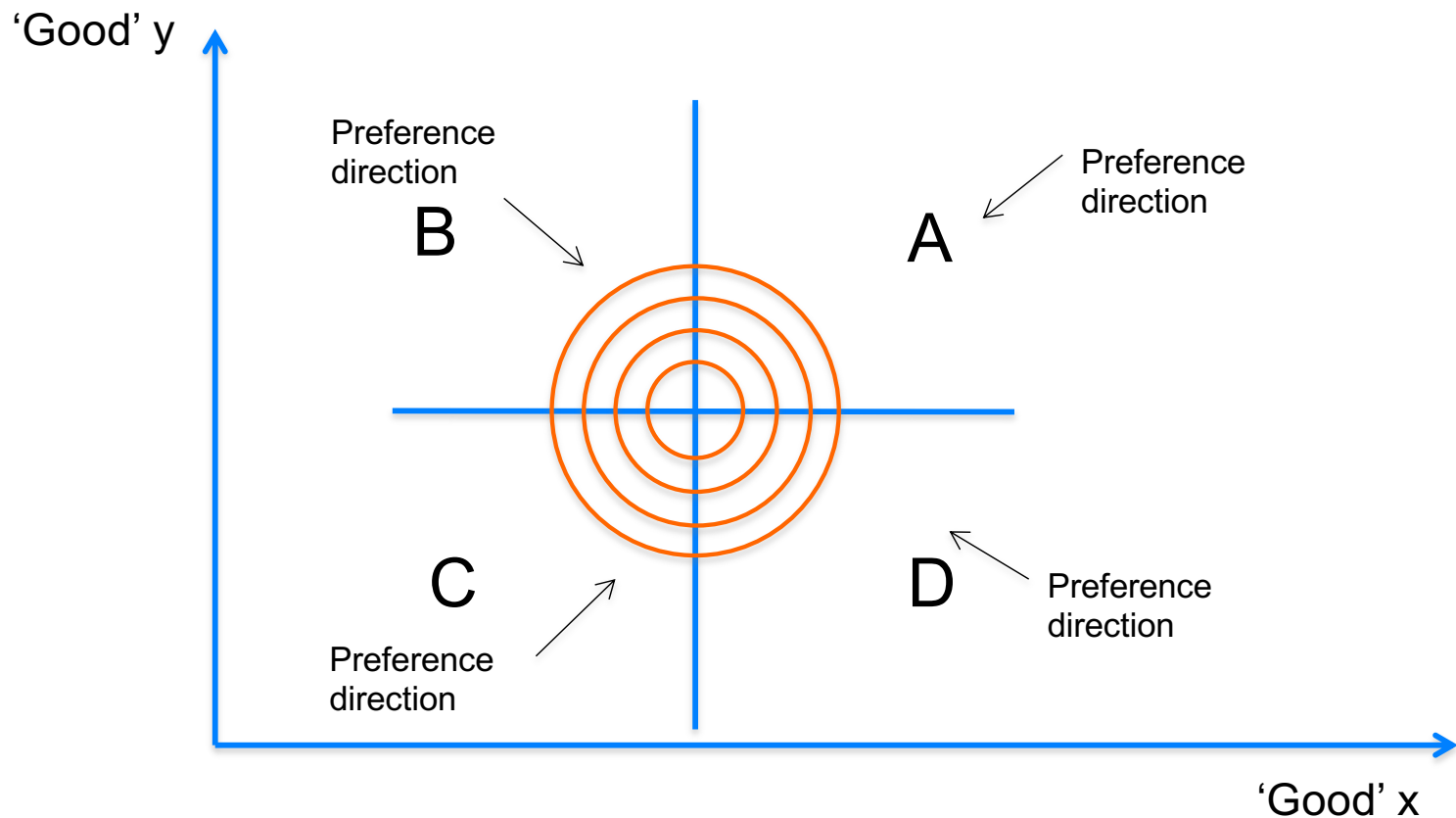
1. A
2. B
3. Not enough information



Good, bads, and neuters

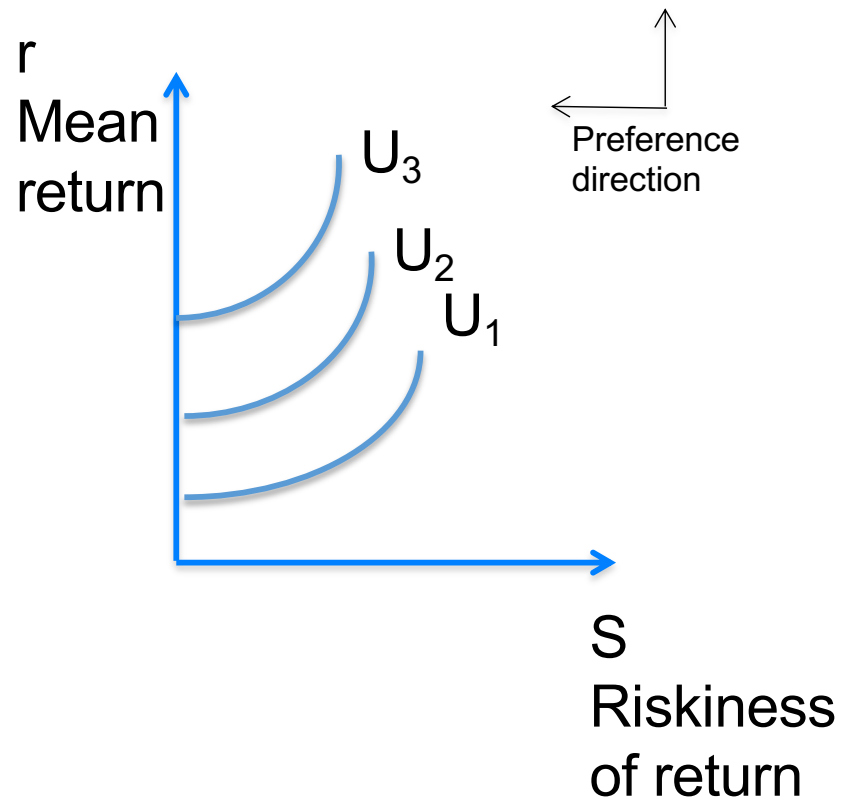
Not all commodities are goods (pollution is a bad, for example)

Also, everyday experience tells us that a commodity can be a good up to a point of satiation, beyond which it becomes a bad



Goods, bads, and neuters

Portfolio preferences



Homework Lecture 3

Draw a indifference curve map of a good (y) and a neuter commodity (x).

HINT: the consumer does not care about having more or less x

Glossary

- **Pursue:** To strive to gain or accomplish
- **Cardinal:** a number denoting quantity
- **Ordinal:** denoting a particular position in a sequence of numbers
- **Thus:** Therefore (example: We have failed. Thus we have to take the consequence)

One-minute paper - Canvas

Write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.